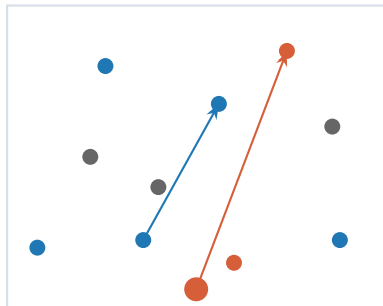


# Effects of Superspreaders in Spread of Epidemic

Spatial SIR model, Monte Carlo simulation, and SARS comparison

Reference: R. Fujie and T. Odagaki, Physica A, 2007

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local and long-range  
transmission paths

# Roadmap

1. Problem and modeling approach
2. Spatial SIR model with two superspreader mechanisms
3. Monte Carlo solving algorithm
4. Reproduced simulation results
5. Sensitivity, robustness, and reference conclusion

# Problem

- ▶ Classical well-mixed SIR models assume similar transmission rates for all infected individuals.
- ▶ Real outbreaks often show strong heterogeneity: a small number of cases cause many secondary infections.
- ▶ The paper models superspreaders directly and asks which mechanism explains spread better.

## Key question

Are superspreaders important because they are more infectious locally, or because they have many social connections?

# Modeling Approach

- ▶ Individuals are fixed in a two-dimensional continuous domain of side length  $L = 10r_0$ .
- ▶ Each individual has state  $S$ ,  $I$ , or  $R$ .
- ▶ Transmission probability depends on periodic distance  $r$  between an infected individual and a susceptible individual.
- ▶ A fraction  $\lambda$  of the population is assigned as superspreaders.

$$\rho = \frac{N}{L^2}, \quad X_i(t) \in \{S, I, R\}.$$

# Two Superspreader Models

## Strong infectiousness model

Superspreaders have higher local infection probability but the same cutoff radius:

$$w_s(r) = \begin{cases} w_0, & 0 \leq r \leq r_0, \\ 0, & r > r_0. \end{cases}$$

## Hub model

Superspreaders have a wider contact radius:

$$w_s(r) = \begin{cases} w_0(1 - r/r_n)^2, & 0 \leq r \leq r_n, \\ 0, & r > r_n, \end{cases}$$

with  $r_n = \sqrt{6}r_0$  for superspreaders.

Both mechanisms are normalized to have comparable expected infection strength; their spatial patterns differ.

# Basic Reproductive Number and Critical Density

$$R_0(\lambda) = \frac{\rho}{w_0} \left[ \lambda \int w_s(r) 2\pi r dr + (1 - \lambda) \int w_n(r) 2\pi r dr \right].$$

$$\rho_c \pi r_0^2 = \frac{R_c}{\lambda + \frac{1-\lambda}{6}}.$$

Strong infectiousness

$$R_c \approx 4.5$$

Hub model

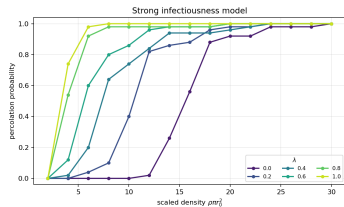
$$R_c \approx 3.2$$

# Monte Carlo Solving Algorithm

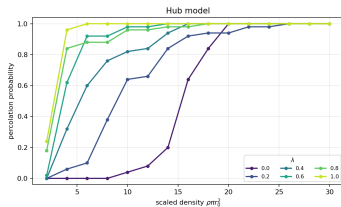
1. Randomly place  $N$  individuals in the domain; set one initial infected individual.
2. Randomly assign superspreader labels using fraction  $\lambda$ .
3. At each step, infected individuals attempt to infect susceptible individuals with probability  $w(r)$ .
4. Newly infected individuals transmit only from the next step.
5. Current infected individuals recover with probability  $\gamma$ .
6. Stop when no infected individuals remain or the maximum number of steps is reached.

Outputs: percolation probability, front distance, propagation velocity, epidemic curve, and secondary infection distribution.

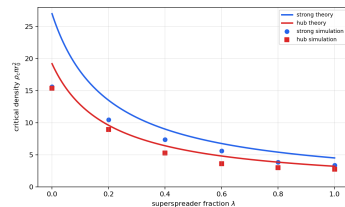
# Result 1: Percolation



Strong



Hub

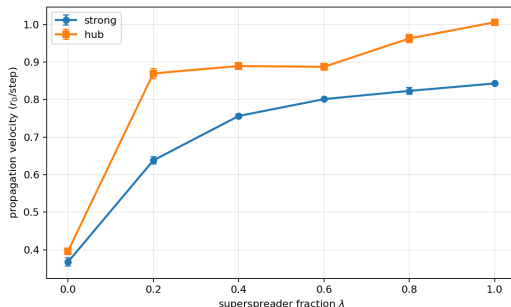
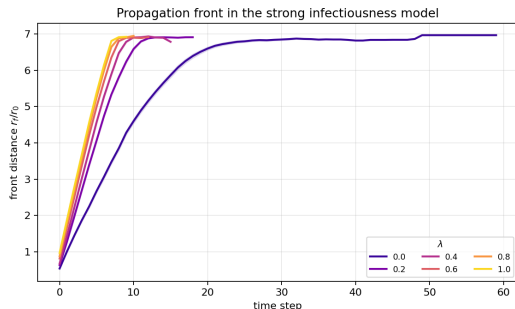


Critical density

Increasing  $\lambda$  lowers the density threshold for percolation. The hub model percolates at lower density.

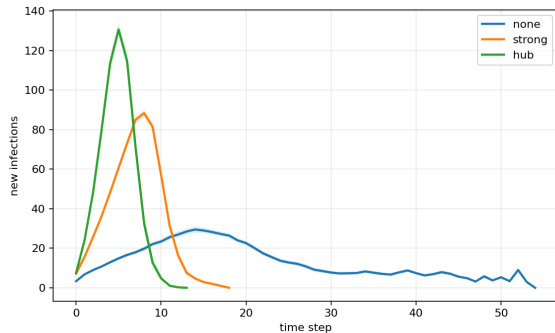


## Result 2: Propagation Speed



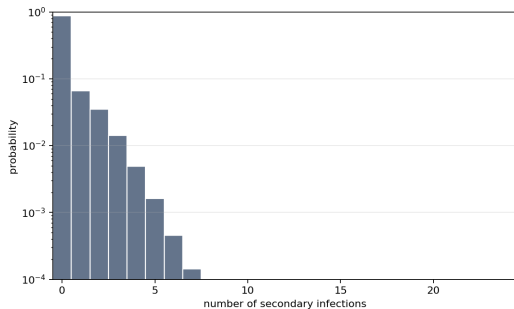
Propagation speed increases with  $\lambda$ . For  $\lambda > 0$ , the hub model spreads faster because it creates long-range infection paths.

# Result 3: Epidemic Curves



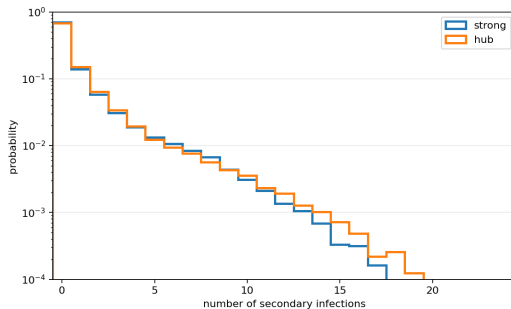
- ▶ No-superspreader outbreaks are broader and peak later.
- ▶ Superspreaders make the peak earlier and sharper.
- ▶ The hub model produces the fastest rise.

## Result 4: Secondary Infections



No superspreaders

Superspreaders create a long-tailed secondary-infection distribution, matching the qualitative pattern discussed in the paper.



With superspreaders,  $\lambda = 0.2$

# Comparison with SARS Data

- ▶ The reference paper compares simulations with SARS in Singapore in 2003.
- ▶ The observed direct secondary cases include several unusually large values, including 12, 21, 23, and 40.
- ▶ The hub model with  $N = 477$ ,  $\rho\pi r_0^2 = 15.0$ ,  $\lambda = 0.4$ , and  $\gamma = 1$  gives the best qualitative epidemic-curve match.

## Reference conclusion

SARS superspreading was better explained by many social connections than by stronger local infectiousness alone.

# Sensitivity and Robustness

- ▶ The model is sensitive to superspreader fraction  $\lambda$ : increasing  $\lambda$  lowers critical density and increases propagation speed.
- ▶ It is sensitive to the mechanism of superspreading: hub contacts produce faster spatial spread than stronger local infectiousness.
- ▶ Population density controls whether infection chains die out or percolate.
- ▶ Recovery probability  $\gamma$  is also important: lower  $\gamma$  means a longer infectious period and more transmission opportunities.

## Robust qualitative findings

Across parameter settings, superspreaders lower the percolation threshold, speed up spread, sharpen epidemic curves, and create long-tailed secondary-infection distributions.

# Limitations

- ▶ Individuals are fixed in space, while real people move and change contacts.
- ▶ Superspreader labels are assigned randomly, but real superspreading depends on behavior, occupation, environment, and biology.
- ▶ The model is best interpreted as a controlled comparison of two mechanisms, not a full prediction tool for every epidemic.

# Conclusion

1. Superspreaders substantially change epidemic dynamics.
2. The hub model spreads faster and has lower critical density than the strong infectiousness model.
3. Long-range social connectivity is the key mechanism in the reference paper's SARS comparison.

## Takeaway

Average transmission rate alone is not enough; epidemic models should include heterogeneity in individual connectivity.